Subspaces

Geometric Algorithms

CAS CS 132

Practice Problem

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 2 & 0 & 2 & 3 \\ 1 & 4 & 0 & 2 \end{bmatrix} \quad \begin{matrix} R_1 \leftarrow R_1 + R_2 \\ R_3 \leftarrow 2R_3 \\ R_2 \leftrightarrow R_3 \\ \longrightarrow \end{matrix} \quad \begin{bmatrix} 3 & 0 & 1 & 5 \\ 2 & 8 & 0 & 4 \\ 2 & 0 & 2 & 3 \end{bmatrix} = B$$

Consider the following pair of matrices A and B which are row equivalent. Find a matrix E such that $EA = B$

Answer

$$R_1 \leftarrow R_1 + R_2$$
$$R_3 \leftarrow 2R_3$$
$$R_2 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 2 & 0 & 2 & 3 \\ 1 & 4 & 0 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 0 & 1 & 5 \\ 2 & 8 & 0 & 4 \\ 2 & 0 & 2 & 3 \end{bmatrix} = B$$

Objectives

- Introduce **subspaces** and **bases**
- Extend our intuitions about planes in $\mathbb{R}^3$ to subspaces in $\mathbb{R}^n$
- Connected subspaces to matrices

Keywords

subspace

closed under addition

closed under scaling

column space

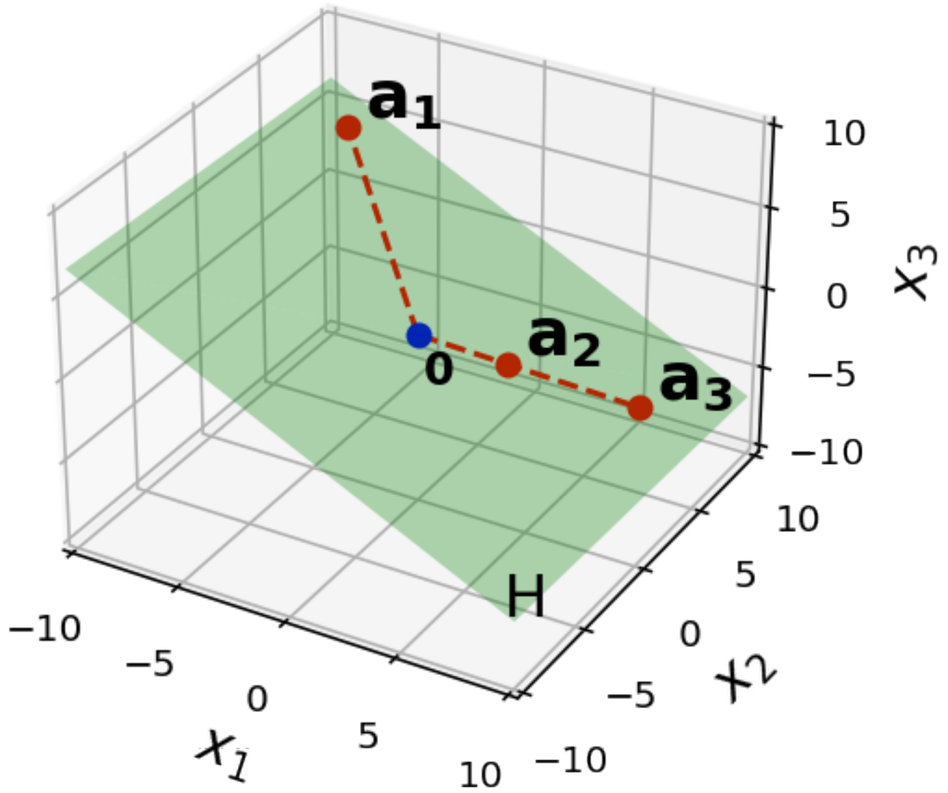
null space

basis

Subspaces

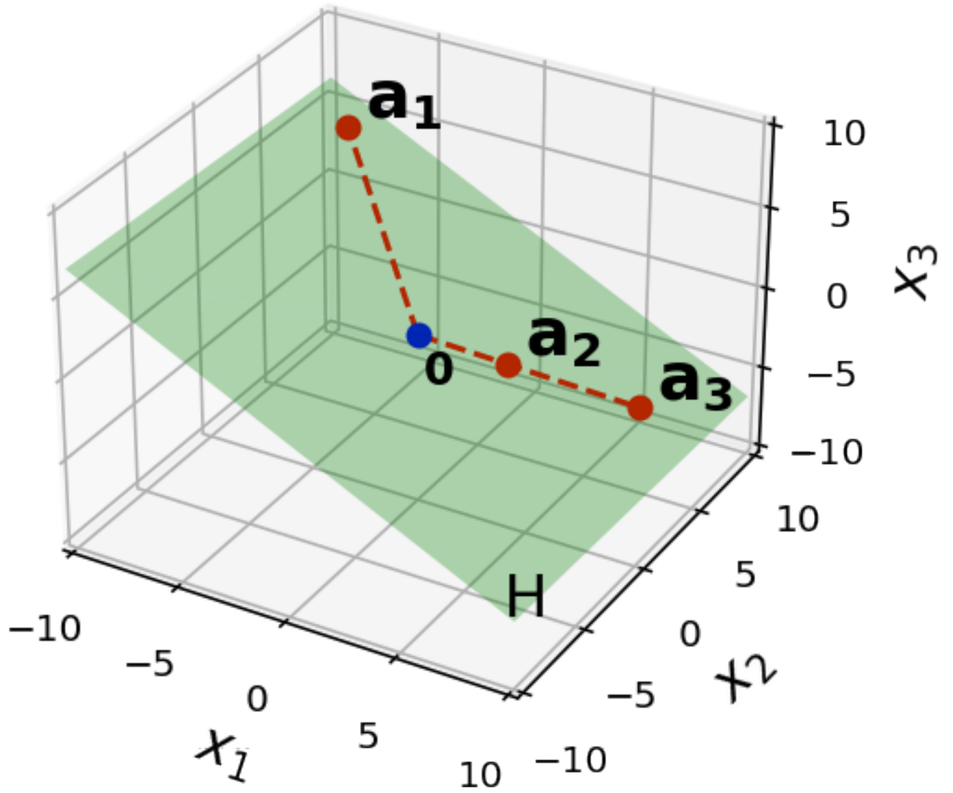
Subspaces

Subspaces


$\mathbf{a_1}$
$\mathbf{a_2}$
$\mathbf{a_3}$
$\mathbf{0}$
H
10
5
0
−5
−10
x_3
10
5
0
−5
−10
x_2
−10
−5
0
5
10
x_1

Subspaces

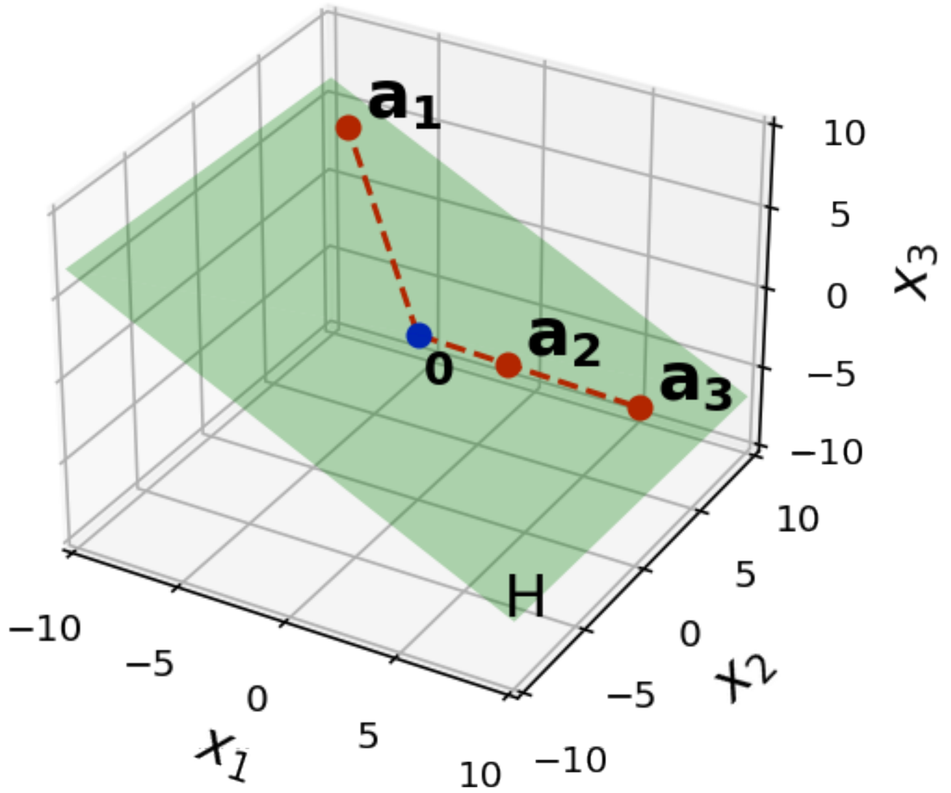
A plane in $\mathbb{R}^3$ looks like a tilted copy of $\mathbb{R}^2$



Subspaces

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Subspaces *generalize* this

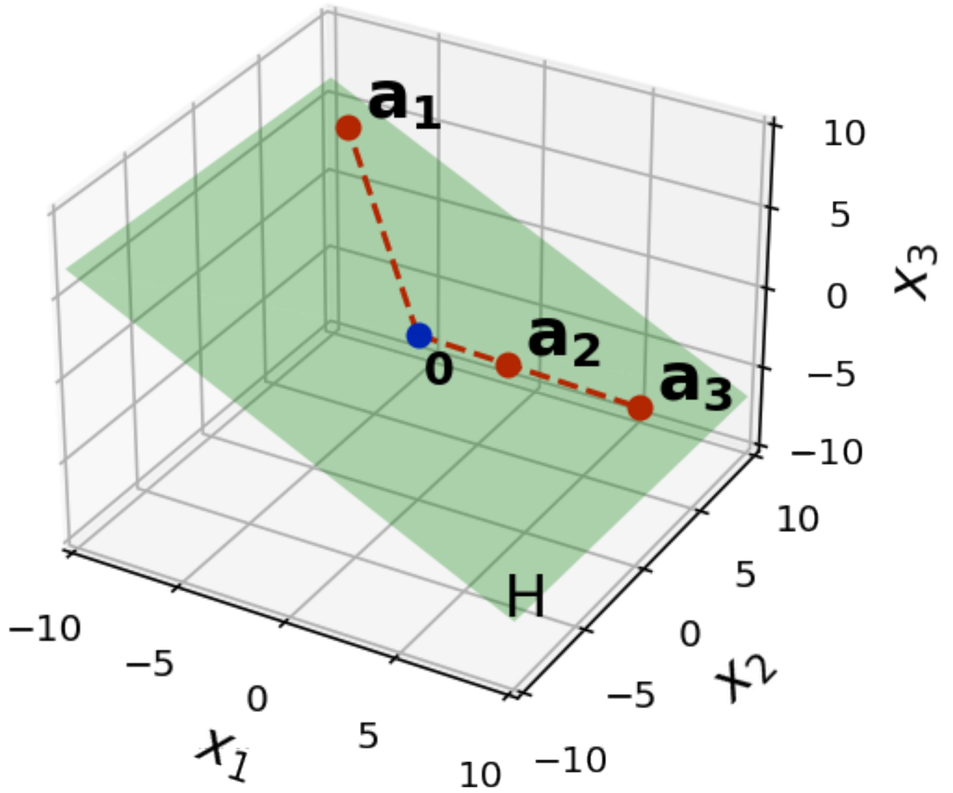


Subspaces

A plane in $\mathbb{R}^3$ looks like a tilted copy of $\mathbb{R}^2$

Subspaces *generalize* this

We can look at "tilted copies" of $\mathbb{R}^7$ sitting in $\mathbb{R}^{11}$



Subspaces (Algebraic Definition)

Subspace (Algebraic Definition)

Df. A **subspace** of $\mathbb{R}^n$ is a set H of vectors in $\mathbb{R}^n$ such that

1. for every $\mathbf{u}$ and $\mathbf{v}$ in H, the vector $\mathbf{u}+\mathbf{v}$ is in H

2. for every $\mathbf{u}$ in H and scalar c, the vector $c\mathbf{u}$ is in H

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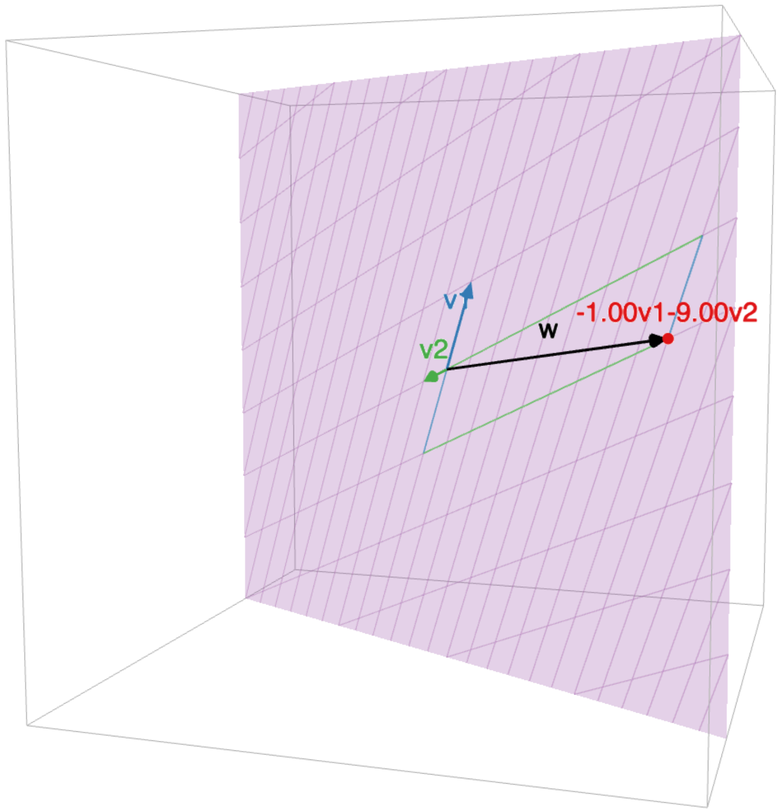
2. for every $\mathbf{u}$ in H and scalar c, the vector $c\mathbf{u}$ is in H — H **is closed under scaling**

!! Subspaces must "live" somewhere !!

How to Think About this

It's not possible to "leave" H by addition or scaling

(recall this is also how we discussed spans)



https://textbooks.math.gatech.edu/ila/spans.html

How To: Verifying Subspaces

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Q. Verify that H is a subspace of $\mathbb{R}^n$

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1. Show if $\mathbf{u}$ and $\mathbf{v}$ are in H then so is $\mathbf{u} + \mathbf{v}$
2. Show if $\mathbf{u}$ is in H then so is $c\mathbf{u}$ for any scalar c

How To: Verifying Non-Subspaces

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Find $\mathbf{u}$ and $\mathbf{v}$ in H where $\mathbf{u}+\mathbf{v}$ is not in H

or

Find $\mathbf{u}$ in H where $c\mathbf{u}$ is not in H for *some* scalar c

Example: Lines in $\mathbb{R}^2$

Subspaces must include the origin

Fact. For any subspace H of $\mathbb{R}^n$, the zero vector is in H. In set notation: $\mathbf{0} \in H$

(this can also be used to verify non-subspace-ness)

Example: The Origin

Fact. The set $\{\mathbf{0}_n\}$ is a subspace of $\mathbb{R}^n$

Example: $\mathbb{R}^n$

Fact. The set $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$

Example: Spans

Fact. For any set of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in $\mathbb{R}^n$ the set $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a subspace of $\mathbb{R}^n$

Subspace in $\mathbb{R}^3$ (Geometrically)

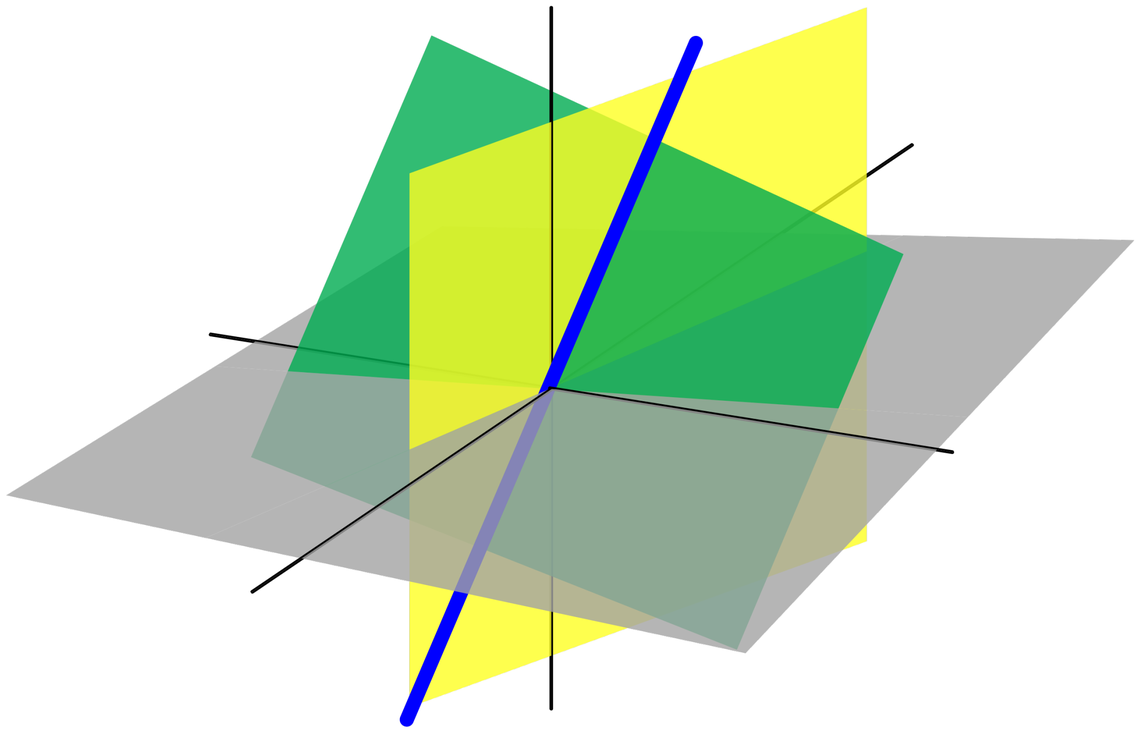
There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. $\{\mathbf{0}\}$ just the origin

2. lines (through the origin)

3. planes (through the origin)

4. All of $\mathbb{R}^3$

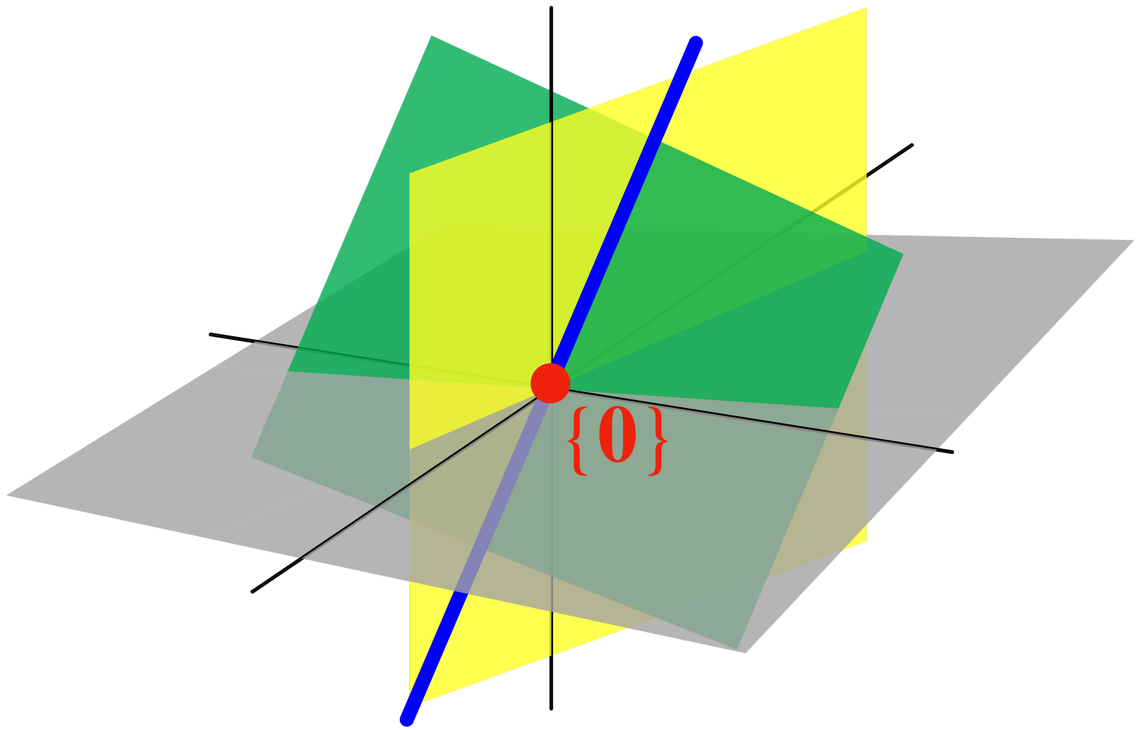


https://commons.wikimedia.org/wiki/File:Linear_subspaces_with_shading.svg

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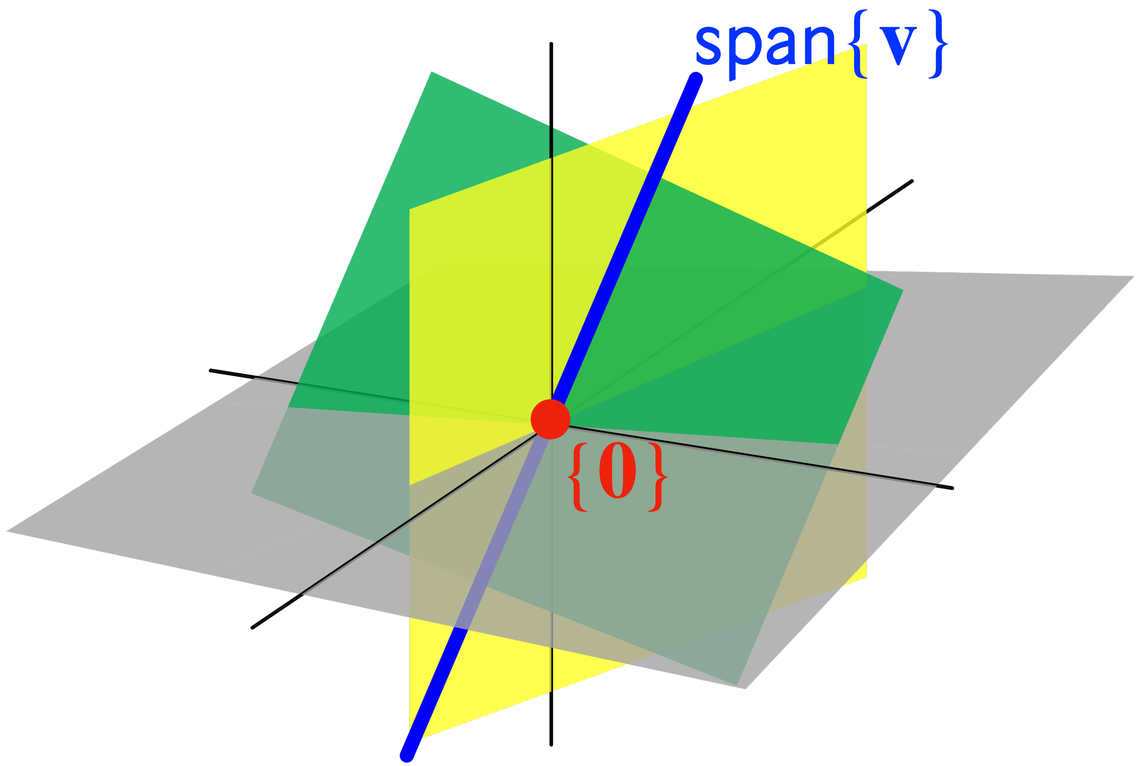


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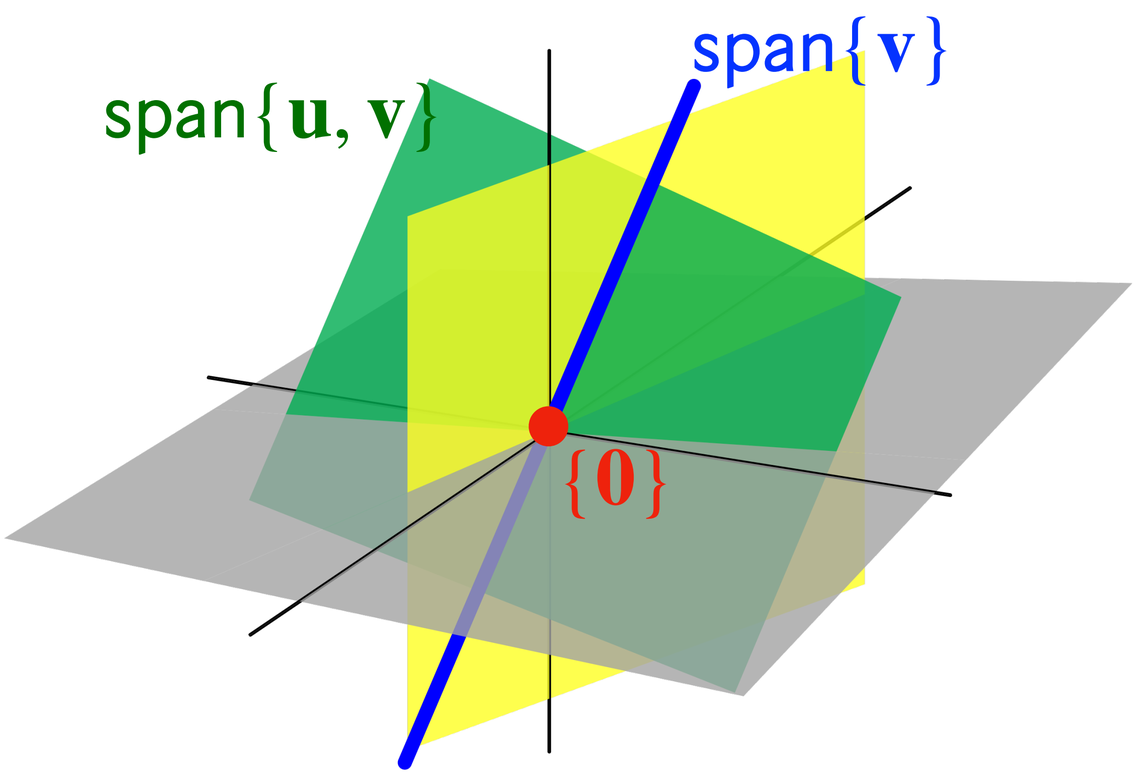


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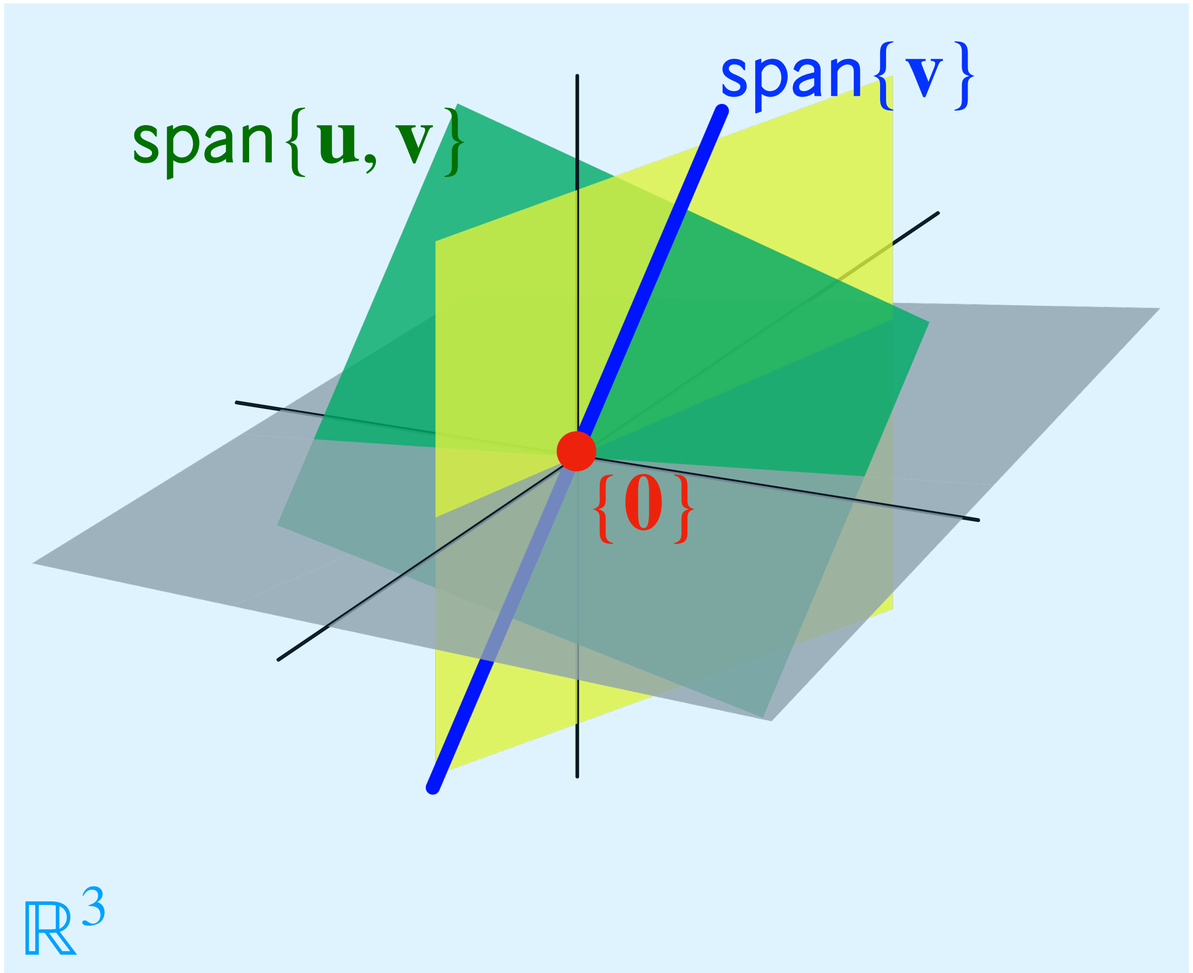


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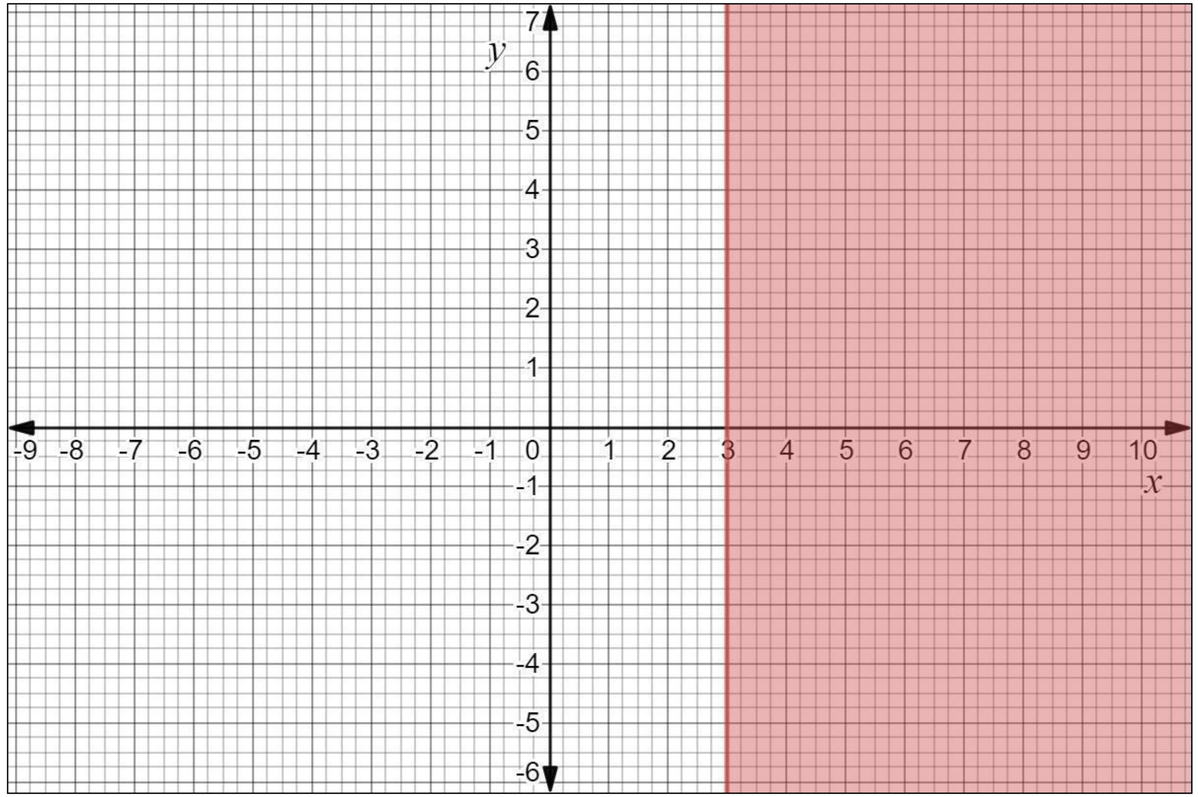
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Non-Example: Bounded Sets

Fact. The set $\{(x, y) : x \geq 3\}$ is *not* a subspace of $\mathbb{R}^2$



https://brainly.com/question/14147114

Question

1. Show that the unit sphere $\{(x, y, z) : x^2 + y^2 + x^2 = 1\}$ *is not a subspace of* $\mathbb{R}^3$

2. Show that the range of a linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$ *is a subspace of* $\mathbb{R}^n$

Answer (1)

Answer (2)

How To: Subspaces and Span

Q. Show that $\mathbf{v}$ lies in the subspace generated by $\mathbf{u}_1, \ldots, \mathbf{u}_k$

A. Show that $\mathbf{v}$ is in $\text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$

We will start using "subspace generated by" and "span of" interchangeably

Subspaces and Matrices

The Connection

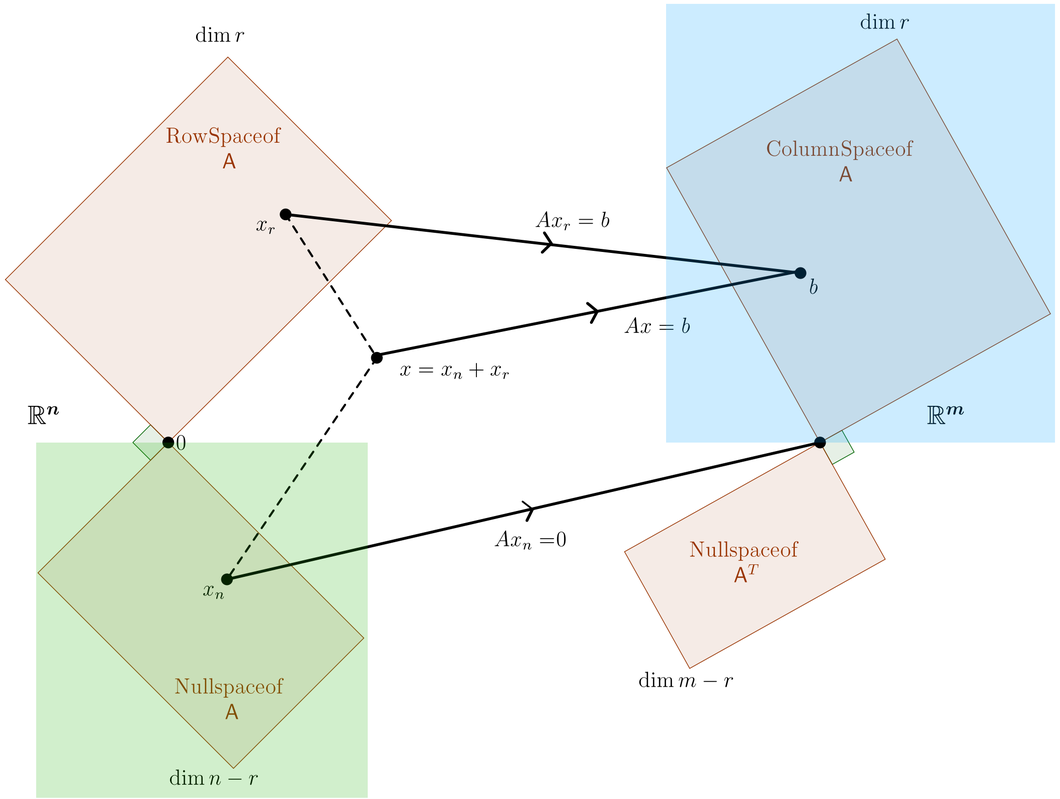
Since matrices can be viewed as:

- collections of vectors
- implementing linear transformations

they have many associated subspaces

Today we'll look at:

- column space
- null space



https://mathworld.wolfram.com/FundamentalMatrixSubspaces.html

Column Space

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Df. *The column space of a matrix is the span of its columns*

Df. *The column space of a matrix is the range of the linear transformation it implements*

Subspace of What?

$$\overset{n}{\overbrace{\begin{matrix}\\ m\end{matrix}\left|\begin{bmatrix} | & | & \dots & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_{n-1} & \mathbf{a}_n \\ | & | & \dots & | & | \end{bmatrix}\right.}}$$

$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots c_n\mathbf{a}_n$ **is a vector in** $\mathbb{R}^m$

$$\text{Col}(A)$$

is a subspace of

$$\mathbb{R}^m$$

Examples

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 & 4 \\ 2 & -2 & 5 \\ 3 & -3 & 6 \end{bmatrix}$$

Null Space

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Df. The **null space** of a matrix A, written $\text{Nul}\, A$, is the set of all solutions to the homogenous equation $A\mathbf{x} = \mathbf{0}$

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The null space of a matrix A is the set of all vectors that are mapped to the zero vector by A

Subspace of What?

$$\underset{m \times n}{A}\ \underset{n \times 1}{\mathbf{v}} = \underset{m \times 1}{\mathbf{0}}$$

n columns

m rows

$\mathbf{v}$ **is a vector in** $\mathbb{R}^n$

$$\text{Nul}(A)$$

is a subspace of

$$\mathbb{R}^n$$

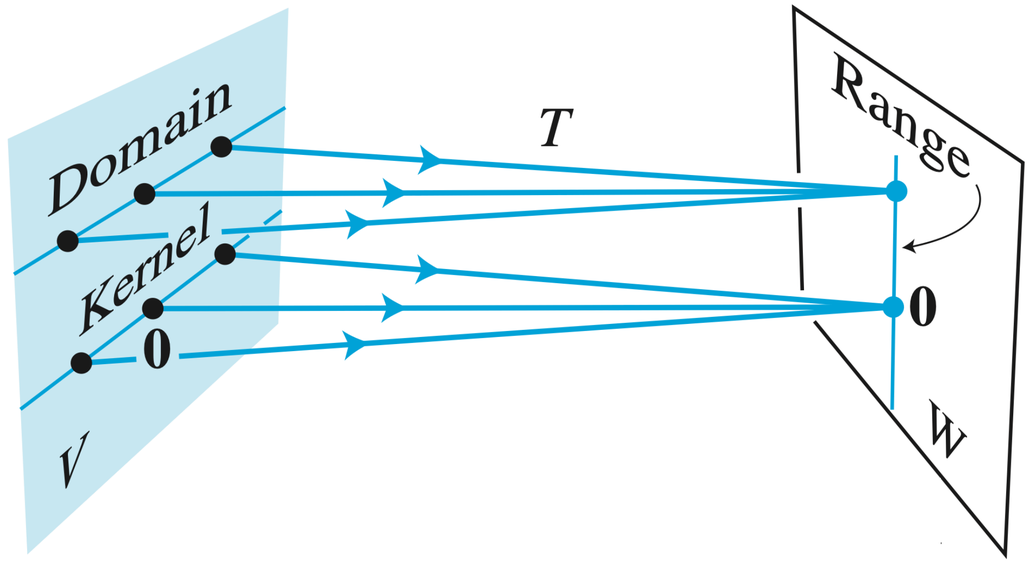
The Null Space is a Subspace

Fact. For any $m \times n$ matrix A, the set $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$

Examples

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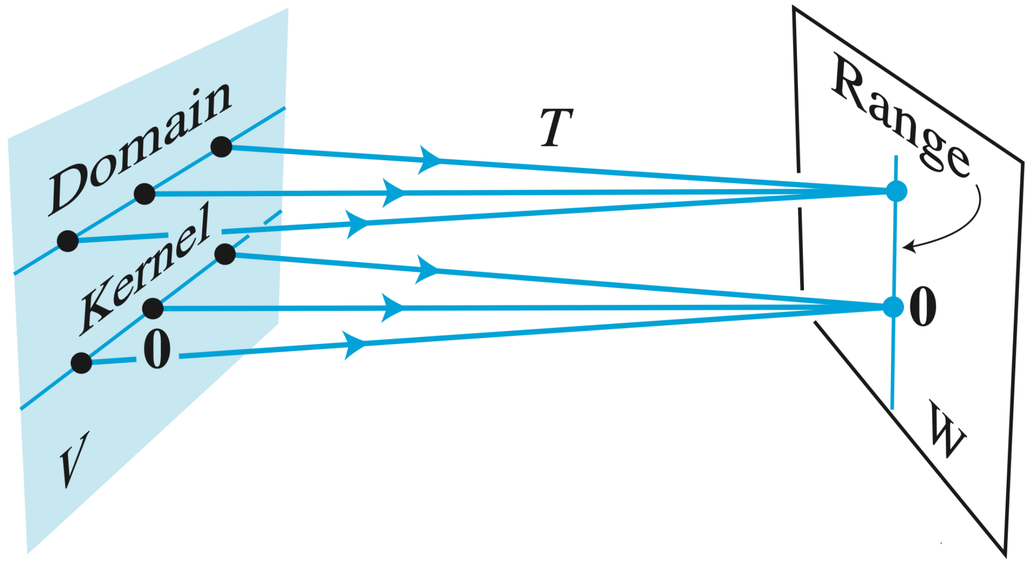
Linear Transformations Perspective



Linear Algebra and its Applications (Lay, Lay, McDonald)

Linear Transformations Perspective

If A implements T then

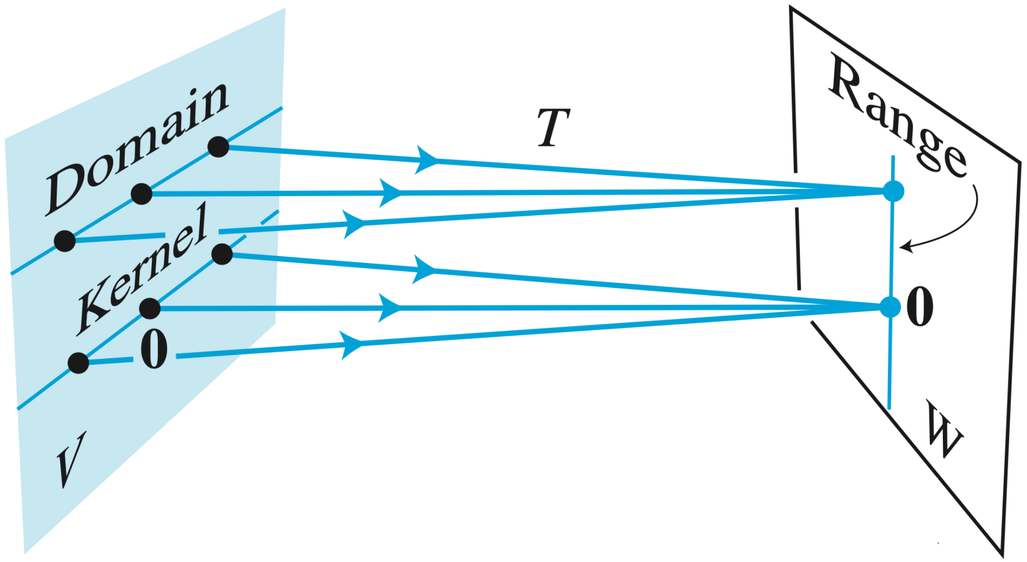


Linear Algebra and its Applications (Lay, Lay, McDonald)

Linear Transformations Perspective

If A implements T then

» $\text{Col}(A)$ is the same as $\text{ran}(T)$, where vectors are "sent" by T

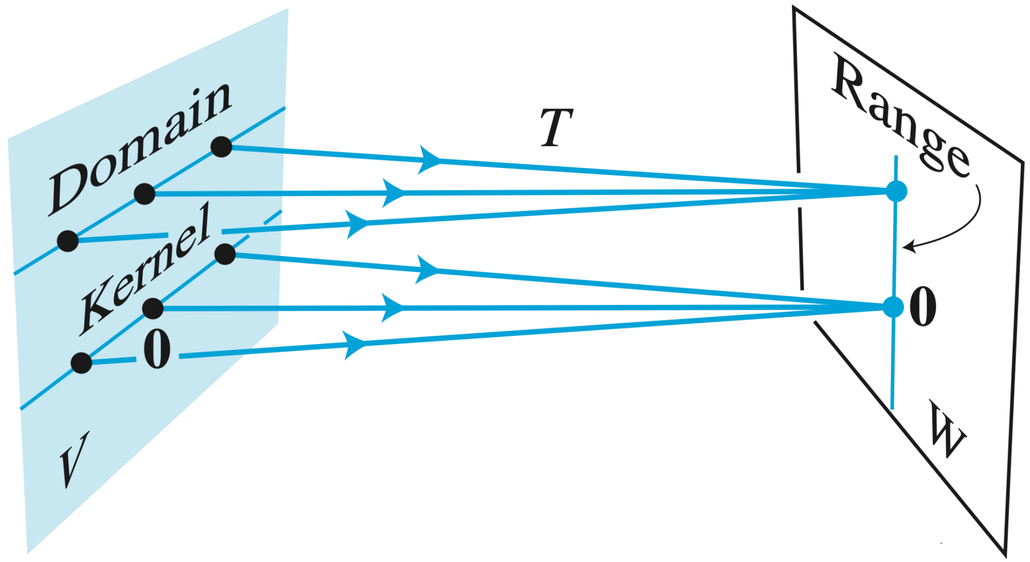


Linear Algebra and its Applications (Lay, Lay, McDonald)

Linear Transformations Perspective

If A implements T then

- $\text{Col}(A)$ is the same as $\text{ran}(T)$, where vectors are "sent" by T
- $\text{Nul}(A)$ is the set of vectors that are "zeroed out" by T, also called the **kernel** of T



Linear Algebra and its Applications (Lay, Lay, McDonald)

Bases

The idea behind a basis

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We've already said spans are subspaces, but the *converse* true too

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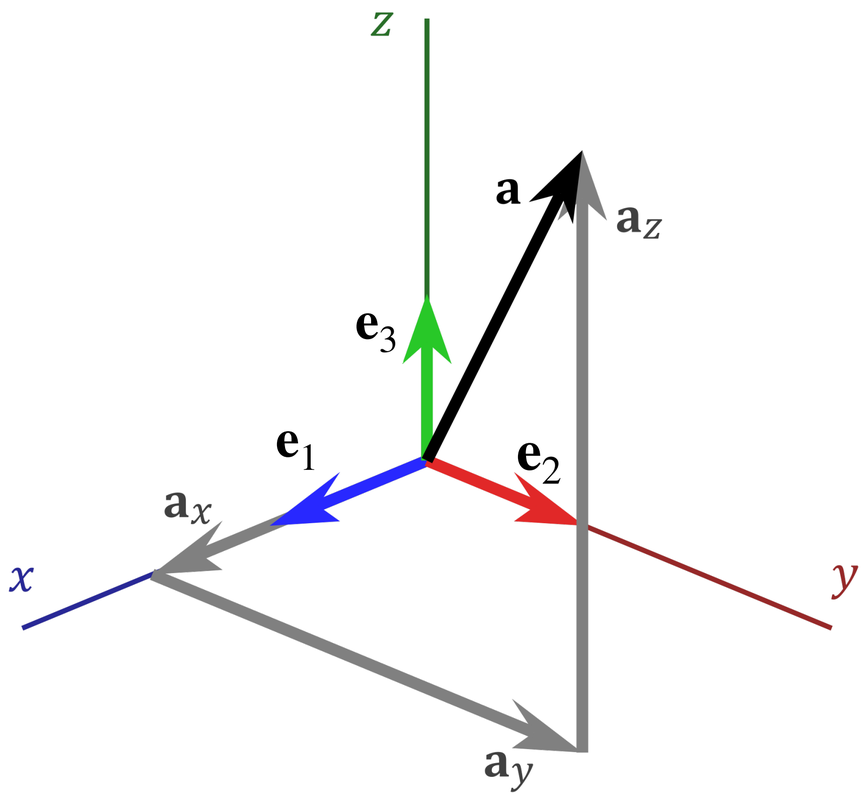
A basis is a "minimal" choice of these vectors, a "concise representation" of a subspace

Recall: Standard Basis

Df. The *n-dimensional standard basis vectors* (or standard coordinate vectors) are the vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ where

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ \vdots \\ i-1 \\ i \\ i+1 \\ \vdots \\ n-1 \\ n \end{matrix}$$

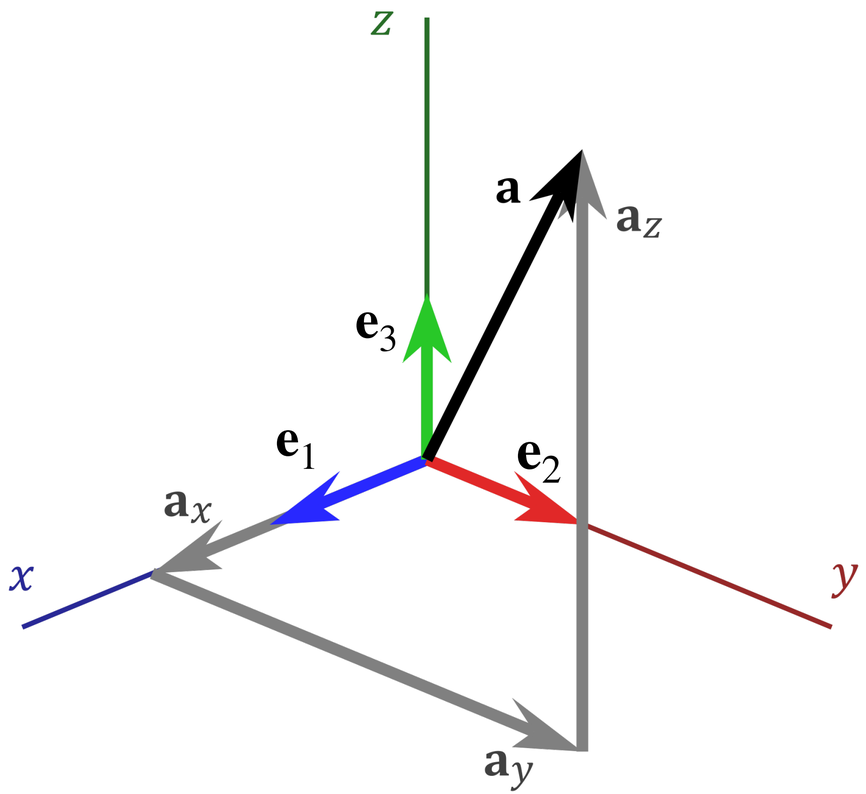
What's special about the standard basis?



What's special about the standard basis?

The n standard basis vectors in $\mathbb{R}^n$:

- are linearly independent
- span all of $\mathbb{R}^n$

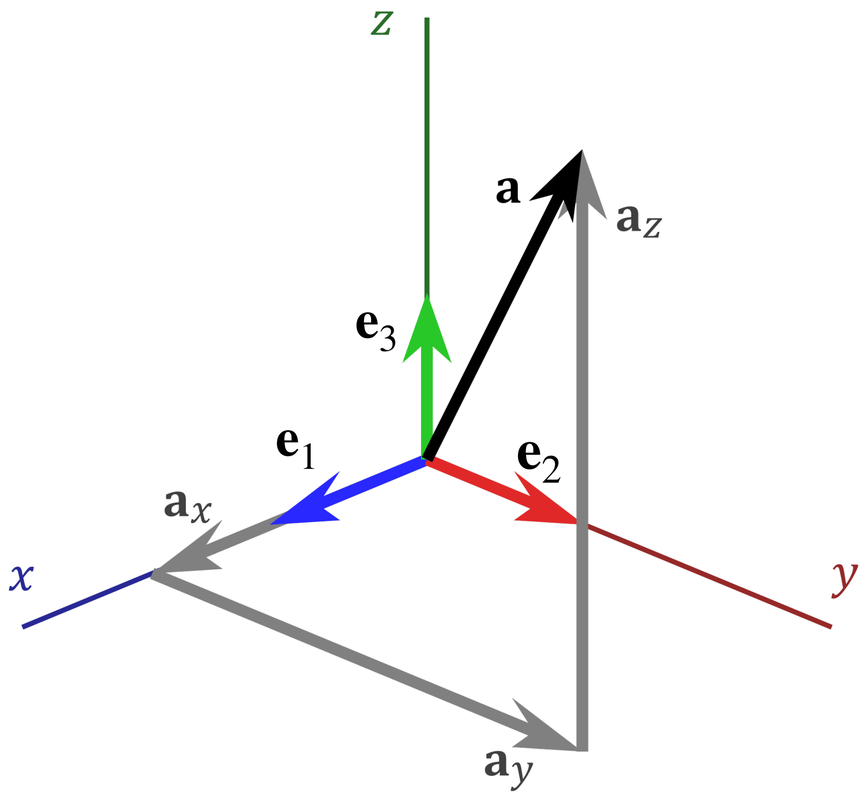


What's special about the standard basis?

The n standard basis vectors in $\mathbb{R}^n$:

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Their span is as "large" as possible while the set of vectors generating the span is as "small" as possible



Basis

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Df. A **basis** for a subspace H of $\mathbb{R}^n$ is a linearly independent set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ of vectors that spans H

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A basis is a *minimal* set of vectors which spans all of H

Example: Standard basis

The standard basis is a basis of $\mathbb{R}^n$

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \ldots + v_n\mathbf{e}_n$$

Column vectors are just weights for a linear combination of the standard basis

Example: Column Space of Invertible Matrices

Fact. The columns of an invertible $n \times n$ matrix form a basis of $\mathbb{R}^n$

Subsets of Spanning Sets

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Thm. If the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ span a subspace H then a subset of them form a basis of H

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How do we do this?

Subsets of Spanning Sets

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We can *remove* vectors from a spanning set until we get a basis

How do we do this?

As usual, by connecting back to matrices

Question

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix} \right\}$$

Is this set of vectors a basis for $\mathbb{R}^3$*?*

Bases of Column Space and Null Space

The Goal of this Last Section

Determine how to find bases for the **column space** and the **null space** of a given matrix

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\text{Nul}(A)$

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\text{Nul}(A)$

The idea. Describe the solutions of $Ax = 0$ as linear combination of vectors

Example

$$A \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Let's write down a general form solution for A:

Parametric Solutions

We can think of our general form solution as a *linear transformation*

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

x_5 is free

$\equiv$

"given values for x_2, x_3, and x_4, I can give you a solution"

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$$\equiv \quad \begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Parametric Solutions

We can think of our general form solution as a *linear transformation*

!! this transformation is only linear !!
!! in the case of homogeneous equations !!

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Example

$$\begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Let's find the matrix implementing this transformation:

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Thm. The columns of this matrix are linearly independent

Example

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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The columns of this matrix *span* $\text{Nul}(A)$

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The columns of this matrix *span* $\text{Nul}(A)$

The columns of this matrix are linearly independent

The columns of this matrix form a basis for $\text{Nul}(A)$

Example

Alternatively, we can write a general form solution so that it is a linear combination of vectors with *free variables as weights*

$$x_1 = 2x_2 + x_4 - 3x_5$$
x_2 is free
$$x_3 = (-2)x_4 + 2x_5$$
x_4 is free
x_5 is free

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\text{Nul}(A)$

A.

1. Find a general form solution for $A\mathbf{x} = \mathbf{0}$
2. Write this solution as a linear combination of vectors *where the free variables are the weights*
3. The resulting vectors form a basis for $\text{Nul}(A)$

An Observation

The *number* of vectors in the basis we found is the same as the number of *free variables* in a general form solution

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

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$\equiv$

$$\begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Practice Problem

$$A \sim \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Determine a basis for $\text{Nul}(A)$

Answer

$$\begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

onto column space...

How To: Finding a basis for the column space

How To: Finding a basis for the column space

Question. Given a $m \times n$ matrix A, find a basis for $\text{Col}(A)$

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How To: Finding a basis for the column space

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We already know the columns of A span $\text{Col}(A)$

So we also already know *some* subset of columns of A form a basis for $\text{Col}(A)$

Which vectors should we choose?

Column Space and Reduced Echelon form

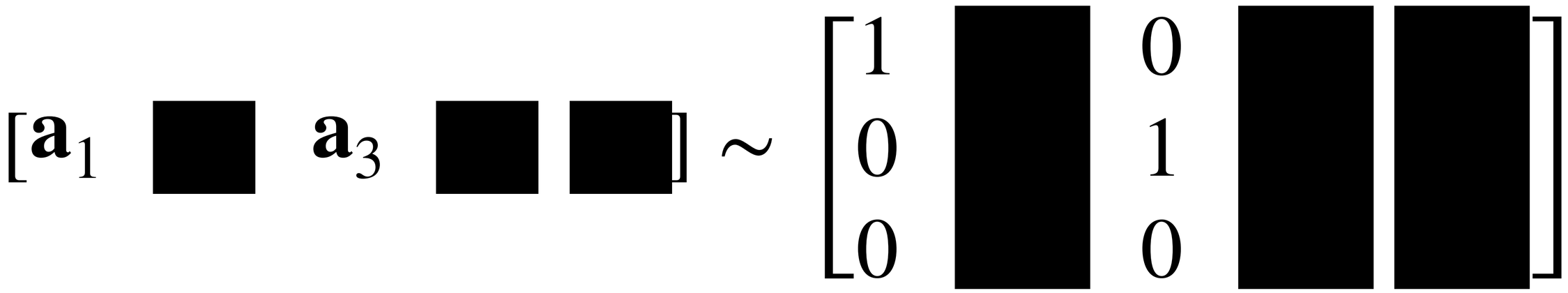
$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column Space and Reduced Echelon form

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The idea. What if we cover up the non-pivot columns?

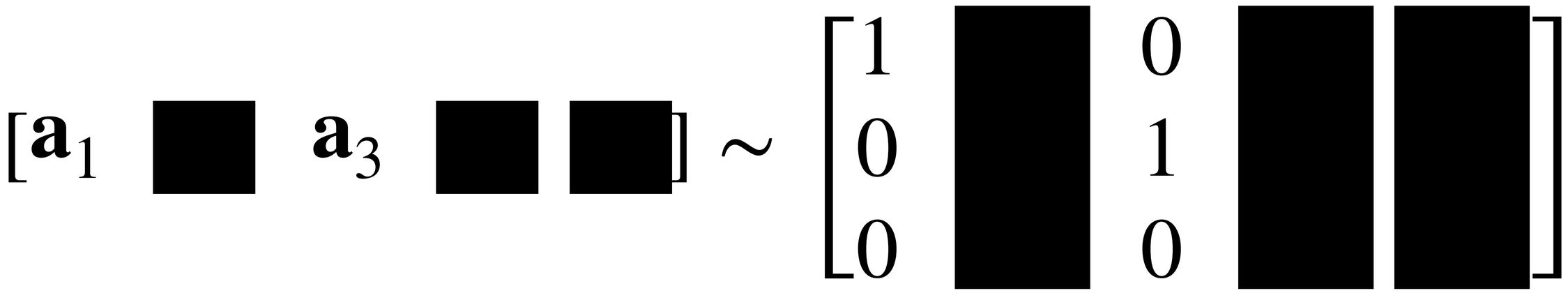
Column Space and Reduced Echelon form



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Then we see $[\mathbf{a}_1 \;\; \mathbf{a}_3]$ has 2 pivots

Column Space and Reduced Echelon form



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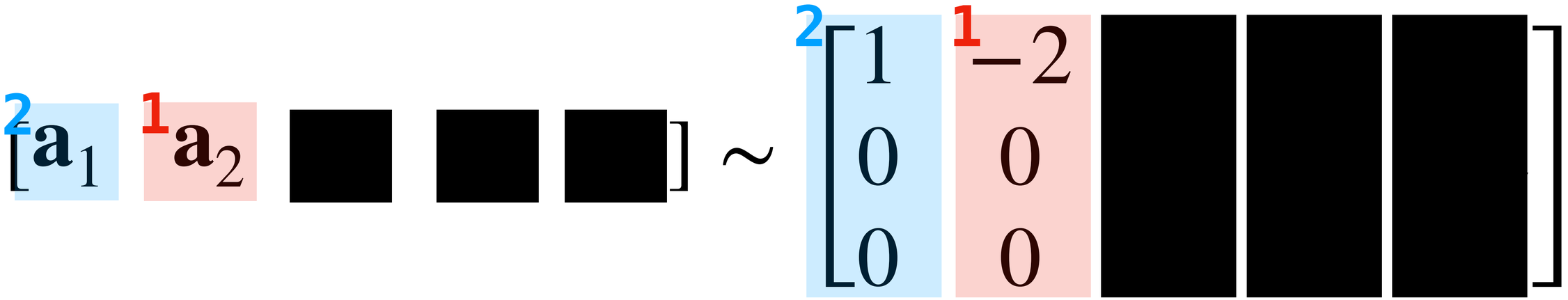
Then we see $[\mathbf{a}_1 \quad \mathbf{a}_3]$ has 2 pivots

So the pivot columns are *linearly independent*

Column Space and Reduced Echelon form

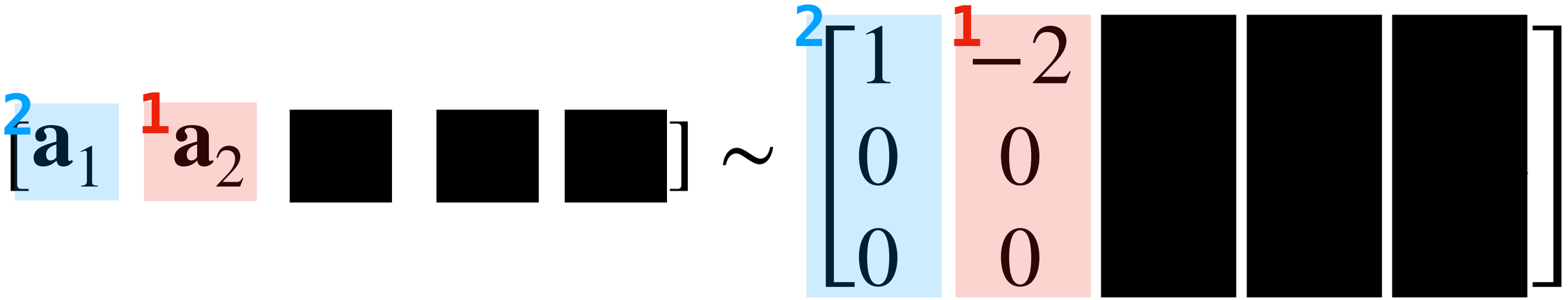
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Column Space and Reduced Echelon form



Observation. $[2\ 1\ 0\ 0\ 0]^T$ is a solution to the system $A\mathbf{x} = \mathbf{0}$

Column Space and Reduced Echelon form



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So $\mathbf{2a}_1 + \mathbf{a}_2 = \mathbf{0}$ and $\mathbf{a}_2 = (-2)\mathbf{a}_1$

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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In general, every non-pivot column of A can be written as a linear combination pivots in front of it

Column Space and Reduced Echelon form

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In general, every non-pivot column of A can be written as a linear combination pivots in front of it

This tells us that $\mathbf{a}_1$ and $\mathbf{a}_3$ *span* Col(A)

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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The takeaway. The pivot columns of A form a basis for $\mathrm{Col}(A)$

Column Space and Reduced Echelon form

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The takeaway. The pivot columns of A form a basis for $\text{Col}(A)$

!! IMPORTANT !!

Choose the columns of A

($\mathbf{e}_1$ and $\mathbf{e}_2$ do not necessarily form a basis for $\text{Col}(A)$)

How To: Finding a basis for the column space

Q. Given a $m \times n$ matrix A, find a basis for $\text{Col}(A)$

A.

1. Find the pivot columns in an echelon form of A
2. The associated columns <u>in A</u> form a basis for $\text{Col}(A)$

Question

$$A = \begin{bmatrix} 1 & -2 & 19 & 0 & -4 \\ 1 & 0 & 9 & 1 & 1 \\ 1 & -1 & 14 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Find a bases for the column space and null space of A

Answer

$$A = \begin{bmatrix} 1 & -2 & 19 & 0 & -4 \\ 1 & 0 & 9 & 1 & 1 \\ 1 & -1 & 14 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Summary

Subspaces define "tilted versions" of $\mathbb{R}^k$ in $\mathbb{R}^n$ (where $k \leq n$)

Bases are compact representation of subspaces as minimal spanning sets

Matrices have useful associated subspaces like the column space and null space